ELEC5305 Project Proposal

A Study of STFT Settings for Speech Signals

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GitHub Project and Pages Links	https://github.com/yasmine-000/elec5305-project-530210546

1. Project Overview

This project will study how different Short-Time Fourier Transform (STFT) settings change the way a speech signal looks and sounds. STFT is a method used to show how the frequency content of a sound changes over time. It is often shown as a spectrogram. In this project, I will use MATLAB to make spectrograms with different settings. I will then change the spectrogram back into a sound signal and compare it with the original recording. The main settings I will test are window length, window type, hop size and FFT size. The aim is to find settings that give a clear spectrogram and also allow the speech signal to be rebuilt correctly.

2. Background and Motivation

A normal Fourier transform shows the frequencies in a whole signal, but it does not show when each frequency appears. Speech changes quickly over time, so this is not enough for studying different parts of a spoken sentence. STFT solves this problem by dividing the signal into short sections. Each section is multiplied by a window and then changed into the frequency domain. The results from all sections are placed together to make a spectrogram.

The length of the window affects the result. A short window can show quick changes in time more clearly, but the frequency details are less clear. A long window gives clearer frequency details, but some fast changes may be harder to see. The amount of overlap between neighbouring sections can also affect how smoothly the signal is rebuilt. The reason to choose this topic is that STFT, windowing and spectrograms are important parts of the first four weeks of ELEC5305. The project will help me understand these ideas through a practical MATLAB experiment instead of only studying the equations.

3. Proposed Method

The project will be completed in MATLAB. I will use several short speech recordings. Some recordings may be made by me, and a small number of public speech recordings may also be used. The recordings will use the same sample rate so that the results can be compared fairly.

First, I will read each audio file into MATLAB. I will divide the signal into short sections and apply the STFT. I plan to compare window lengths of 256, 512, 1024 and 2048 samples. I will also compare common windows such as rectangular, Hann and Hamming windows. Two overlap settings will be tested: 50% overlap and 75% overlap. I will use a few FFT sizes to see how they change the display of the spectrum.

For each setting, the program will plot a spectrogram and listen to the rebuilt sound. The program will compare whether speech details, harmonics and short changes can be seen clearly. The program will then use the inverse STFT and overlap-add method to rebuild the signal. The rebuilt signal will be compared with the original signal. A simple error value, signal-to-noise ratio and

processing time will be used for the comparison. These tests will show which settings work well and which settings cause larger errors. The MATLAB code will include comments so that each step is easy to understand.

4. Expected Outcomes

At the end of the project, a working MATLAB program, which can read a speech recording, create a spectrogram and rebuild the sound, is expected. The program will allow the user to change the window length, window type, hop size and FFT size. The results are expected to show the difference between short and long windows and the effect of using more overlap. The final report will include spectrogram pictures, a simple table of results and a discussion of which settings are most suitable for speech signals. The GitHub repository will contain the MATLAB code, the project proposal, example results and a README file explaining how to run the program.

5. Timeline

Weeks	Task
1-2	Choose the topic and make the project plan.
3-5	Read related papers and collect speech recordings.
6-7	Write the MATLAB STFT code and create spectrograms.
8-9	Rebuild the signals and compare different settings.
10-11	Check the results and improve the MATLAB code.
12-13	Finish the final report, README and final project files.

6. References

[1] J. B. Allen and L. R. Rabiner, “A unified approach to short-time Fourier analysis and synthesis,” Proceedings of the IEEE, vol. 65, no. 11, pp. 1558-1564, 1977.

[2] F. J. Harris, “On the use of windows for harmonic analysis with the discrete Fourier transform,” Proceedings of the IEEE, vol. 66, no. 1, pp. 51-83, 1978.

[3] D. W. Griffin and J. S. Lim, “Signal estimation from modified short-time Fourier transform,” IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 32, no. 2, pp. 236-243, 1984.

[4] L. R. Rabiner and R. W. Schafer, Theory and Applications of Digital Speech Processing. Pearson, 2011.